

Weighted Automata Learning with a Hard Inductive Bias

A weighted finite automaton (WFA) over a semiring $\mathbb{S}$ is a quintuple $\alpha = \langle Q, \Sigma, \delta, I, F \rangle$, where Q is a finite set of states, Σ is a finite alphabet, $\delta: Q \times \Sigma \times Q \rightarrow \mathbb{S}$ is the transition weight function, $I: Q \rightarrow \mathbb{S}$ is the initial weight function, and $F: Q \rightarrow \mathbb{S}$ is the final weight function. Primarily, we deal with $\mathbb{B}$ or $\mathbb{R}$ semirings. Likewise, a context-free grammar (CFG) is a quadruple $G = \langle V, \Sigma, P, S \rangle$ where V is a set of nonterminals, $P \subset V \times (V \cup \Sigma)^+$ is a set of productions, and $S \in V$ is a distinguished start symbol. We use $\mathcal{L}(\cdot)$ to denote the language these devices generate.

In this work, we assume sample access to a language model which has learned a distribution $P_\theta(\sigma)$ with support over $\sigma: \Sigma^+$ but largely concentrated within $\mathcal{L}(G)$, where G is a known grammar. *Our goal is to distill a surrogate WFA, α_θ^* , such that for all $\sigma \in \mathcal{L}(G)$, $P_\theta(\sigma)$ approximates the learned distribution, $P_\theta(\sigma \mid \sigma \in \mathcal{L}(G))$, while preserving strict language inclusion ($\mathcal{L}(\alpha^*) \supset \mathcal{L}(G)$), where $\alpha^* \equiv_{\mathbb{B}} \alpha_\theta^*$.* This problem has applications to constrained decoding and program repair.

Given an irregular CFG, G , there is an infinite descending chain, $\mathcal{L}(\alpha_0) \supset \dots \supset \mathcal{L}(\alpha_n) \supset \mathcal{L}(G)$. Within this hierarchy, Nederhof (2000) defines a regular approximation, $\mathcal{L}(\alpha_d) \supset \mathcal{L}(G)$, whereby every word $\sigma \in \mathcal{L}(\alpha_d)$ consists of a sequence of terminals witnessed by some path through a recursive transition network (RTN) that does not enforce balanced recursion beyond a desired maximum call stack depth, d . For a detailed presentation, we refer the reader to Mohri and Nederhof (2001).

Early work from Dill et al. (1991) imposes a simulation preorder to guarantee language inclusion, but assumes α and G to both be regular. Two main approaches to WFA learning are most pertinent: Bailly et al. (2009) and Balle et al. (2014) develop a spectral framework that provides a theoretically optimal low rank approximation, but assumes an empirical sample, $\hat{\mathbf{H}}$, from an infinite-dimensional Hankel matrix, $\mathbf{H}$. The second approach, from Weiss et al. (2019) proposes $\mathbf{WL}^*$, an adaptation to Angluin’s $\mathbf{L}^*$ algorithm which learns a WFA from queries and counterexamples. More recently, Pasti et al. (2024) generalizes WFA learning to arbitrary semifields and unifies both spectral and $\mathbf{L}^*$ learning. Existing work on WFA learning does not explicitly consider strict language inclusion.

Concretely, we propose a simple counterexample-guided abstraction refinement (CEGAR) loop that iteratively refines an initial automaton until a desired precision or $|Q|_{\max}$ threshold is reached:

1. Initialize $\alpha'_\theta \leftarrow \alpha_d$ using Nederhof’s d -RTN construction, weighted via global normalization.
2. Propose a refinement $\alpha_\theta \preceq \alpha'_\theta \iff D_{\text{KL}}(P_\theta \parallel \alpha_\theta) < D_{\text{KL}}(P_\theta \parallel \alpha'_\theta)$ and $\alpha_\theta \sqsubseteq_G^m \alpha'_\theta$ (see Fig. 1).
3. Binarize α_θ and verify language inclusion $\mathcal{L}(\alpha) \supset \mathcal{L}(G)$ holds by checking $\emptyset = \mathcal{L}(G) \cap \overline{\mathcal{L}(\alpha)}$.
4. If verification fails, collect witnesses $x \in \mathcal{L}(G) \cap \overline{\mathcal{L}(\alpha)}$, otherwise proceed directly to step 2.
5. Use witnesses to repair α until inclusion is restored, renormalize as α'_θ , then go to step 2.

This procedure yields a WFA that structurally over-approximates the target language, $\mathcal{L}(G)$, and distributionally approaches $P_\theta(\sigma)$, our language model’s learned distribution. One may further consider parsimony, description length and other model selection criteria to stabilize convergence.

It may be possible to forgo Nederhof’s d -RTN initialization and learn a regular approximation by spectral decomposition, increasing parameter efficiency at the cost of more expensive training. In future work, it would be interesting to relax the hard constraint with a soft inductive bias that incorporates a surrogate loss, à la Wilson (2025), bypassing the need for a CEGAR loop entirely. Enriched models, such as the context-free family are also conceivable; for instance, Considine (2025) considers a context-free approximation to a programming language type system, a technique which could be adapted to learning a probabilistic CFG that over-approximates a mildly context-sensitive language. Experimental results will appear in a forthcoming FLaNN poster, pending acceptance.

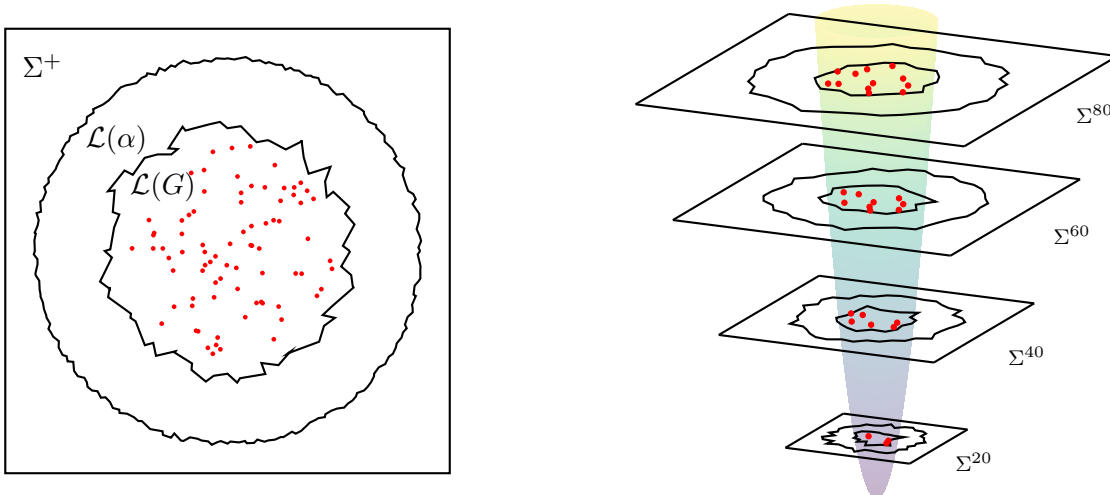


Figure 1: One can measure the tightness of a sound over-approximation by considering finite models,

$$\rho(\alpha, G, m) = \frac{\sum_{n=1}^m |\mathcal{L}(G) \cap \Sigma^n|}{\sum_{n=1}^m |\mathcal{L}(\alpha) \cap \Sigma^n|}, \text{ which induces an order } \alpha \sqsubseteq_G^m \alpha' \iff \rho(\alpha', G, m) < \rho(\alpha, G, m).$$

where ρ is calculable approximately by Boltzmann sampling or analytically via generating functions.

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